

Thyroid Hormone Physiology and Regulation

Thyroid hormones act on target tissues by activating cytoplasmic receptors that subsequently translocate to the nucleus and activate specific thyroid-responsive genes. Thyroid hormone synthesis is heavily dependent on iodine, and changes in iodine intake can result in either hyperthyroidism or hypothyroidism.

Regulation of Thyroid Hormone Synthesis

Synthesis of thyroid hormone is regulated by the pituitary gland through the release of **thyroid-stimulating hormone (TSH)**, which is itself stimulated by **thyrotropin-releasing hormone (TRH)** from the hypothalamus. TSH acts on the thyroid gland by binding to a G-protein-coupled receptor (TSH-R), triggering thyroid hormone synthesis and release.

Aberrant activation of TSH-R underlies the hyperthyroidism seen in **Graves disease**, where nearly all patients have **long-acting thyroid stimulator (LATS) autoantibodies** that bind and activate the TSH-R. TRH also acts via a G-protein-coupled receptor, and administration of recombinant TRH followed by measurement of TSH response can help distinguish between primary (thyroidal) and central (hypothalamic-pituitary) causes of hypothyroidism.

Feedback Regulation

Circulating levels of **thyroxine (T_4)** and **triiodothyronine (T_3)** tightly regulate the release of both TSH and TRH through negative feedback mechanisms. Persistent stimulation of the thyroid gland can lead to **goiter**, a hypertrophy of the gland.

Thyroid Hormone Transport and Availability

The majority of thyroid hormones in the blood are bound to plasma proteins: - **Thyroid-binding globulin (TBG)** – the primary binding protein - **Transthyretin** (formerly prealbumin) - **Albumin**

Various medical conditions and medications can alter the levels of these binding proteins or affect the peripheral conversion of T_4 to T_3 —the biologically active form responsible for most of thyroid hormone's effects. These changes can influence tissue availability of thyroid hormone independently of thyroid synthesis.

Thyroid hormone levels may also be affected by several medications, including those commonly prescribed by dermatologists.

Classification of Hyperthyroidism

Hyperthyroidism can be classified based on **radioiodine uptake**: - **High uptake**: indicates increased thyroid hormone synthesis (e.g., Graves disease, toxic multinodular goiter) - **Low or absent uptake**: suggests either an extrathyroidal source of hormone or destruction of thyroid tissue with release of preformed hormone (e.g., thyroiditis)

Genetic predisposition and environmental factors, such as infection, are believed to contribute to the development of autoimmune thyroid diseases.

In **toxic adenoma**, molecular studies have identified frequent activating mutations in the **TSH-R**, leading to constitutive (TSH-independent) stimulation of thyroid hormone production.

Cutaneous Manifestations of Thyroid Disease

While some skin findings in hyperthyroidism are indirect, most cutaneous manifestations result from the **direct effects of thyroid hormones** on: - Keratinized epithelium (epidermis, hair, nails) - Stromal cells in the dermis

Thyroid hormones are essential for optimal **epidermal proliferation** in both in vitro and in vivo settings. They regulate genes involved in **keratinocyte**

proliferation and stimulate the expression of specific **keratins**, influencing skin, hair, and nail health.